Exact Haar Entropy in the Local $SU(2)$ Partition Comparison

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Abstract

We isolate the exact geometric entropy paid by small comparison sets in $SU(2)$. The result removes an ambiguity in the conditional partition-ratio frontier: a local denominator ball has cubic, rather than Gaussian, Haar mass, and its negative logarithm must be paid by the available action gap. The statement is finite-dimensional and exact; it does not bound the complete Yang–Mills numerator.

Equip $SU(2) \cong S^3$ with its bi-invariant geodesic metric and normalized Haar probability μ_H. Write $B_r(g) = \{h : d(g,h) \le r\}$ and $v(r) = \mu_H(B_r(g))$, which is independent of g.

Theorem 1 (Exact ball and shell mass)**.** *For $0 \le r \le \pi$,*

$$v(r) = \frac{r - \sin r \cos r}{\pi} = \frac{2}{\pi}\int_0^r \sin^2 u \, du, \qquad v'(r) = \frac{2\sin^2 r}{\pi}.$$

Consequently a shell $a \le d(g,h) \le b$ has normalized mass $v(b) - v(a)$.

Proof. Geodesic polar coordinates on the unit three-sphere have volume element $4\pi \sin^2 u \, du$, while $\mathrm{vol}(S^3) = 2\pi^2$. Division by the total volume gives the integral. Integration and differentiation give the two displayed formulas. Bi-invariance removes the center g. □

Theorem 2 (Certified small-ball entropy)**.** *For $0 \le r \le 1$,*

$$\frac{2r^3}{3\pi}\left(1 - \frac{r^2}{5}\right) \le v(r) \le \frac{2r^3}{3\pi}.$$

For $0 < r \le 1$ this is equivalently

$$\log\frac{3\pi}{2r^3} \le -\log v(r) \le \log\frac{3\pi}{2r^3} - \log\left(1 - \frac{r^2}{5}\right).$$

Proof. On $[0,1]$, $u - u^3/6 \le \sin u \le u$. Hence $u^2(1 - u^2/3) \le \sin^2 u \le u^2$; the omitted $u^6/36$ term is nonnegative. Integration in the exact formula gives the volume bounds. The logarithm is order preserving and $-\log$ reverses the volume inequalities as displayed. □

Corollary 1 (Two-center atlas)**.** *The radius-$\pi/2$ balls about I and $-I$ cover $SU(2)$. Each has Haar mass $1/2$; their intersection is the equatorial two-sphere and has Haar measure zero. The atlas therefore has multiplicity one almost everywhere and two on the common boundary.*

Meaning for the partition-ratio programme

At the Stage 25 local radius $r = 0.00614955590043087$, the exact denominator entropy $-\log v(r)$ is recorded by the accompanying executable certificate. The leading behavior is

$$-\log v(r) = 3\log(1/r) + \log(3\pi/2) + O(r^2).$$

Thus shrinking a comparison ball improves local Taylor control while charging three logarithmic entropy units per radial decade. Any proof based on a denominator restriction must retain this cost. A pointwise action minimum, without its integration volume and boundary-state normalizer, is insufficient.

Remark 1 (Claim boundary)**.** *The theorem proves Haar geometry only. It does not compare the nonlinear good partition uniformly over retained boundary states, estimate the charged bad numerator, establish volume-uniform basin entry, iterate an RG map, or construct a continuum Hamiltonian.*